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Course Name - Data Analytics With GenAI

Subject - Tableau

Topic - Calculated Fields, Parameters & LOD Expressions

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Tableau Workbook - Link

Theoretical Questions :

Question 1 : What is a Calculated Field and why is it important in business dashboards?

Answer 1:

A Calculated Field is a new column you can create inside Tableau based on a formula. It's not present in your original data set, but it's calculated on the fly.

Example: If you have a data set with columns for Sales and Cost, you can create a calculated field called Profit % with this formula:

([Sales] - [Cost]) / [Sales] * 100

Why it's important:

You do not have to go back and alter your original data set

It allows you to create business metrics like profit percentage, delivery classification, etc.

It makes your dashboards dynamic and meaningful for your managers to view your KPIs instead of just numbers.

Question 2 : Why can't parameters change visuals directly in Tableau?

Answer 2:

A "Parameter" is simply an input value (e.g., number, text, date) that the end user can change. The parameter is simply holding the value, and it doesn't have any "logic" built in.

In order to actually affect the visual, the parameter must be "attached" to a "Calculated Field" or "Filter" that uses the parameter's value.

Analogy: The parameter is like the volume control on the TV's remote control. If you turn the volume control, nothing will happen unless the TV is "programmed" to respond to the control. The calculated field is the "programming."

Example:

Parameter: Top N = 5 (user input)

Calculated Field: RANK(SUM([Sales])) <= [Top N] (this calculated field "uses" the parameter)

Question 3 : When should EXCLUDE LOD be used?

Answer 3:

EXCLUDE LOD is used when you want to exclude one dimension from the current view's level of detail.

Use it when: You have your view divided into multiple dimensions, and you want to ignore one dimension for one of the metrics.

Example: You have a view that shows Sales divided into City AND Category. You want to have another column that shows "Total Sales for the City, ignoring Category." You would use:

{ EXCLUDE [Category] : SUM([Sales]) }

You are getting city-level total even though the view is at City AND Category level.

Simple rule: Use EXCLUDE when you want to "zoom out" from the current level of detail.

Question 4 : Difference between FIXED and INCLUDE LOD expressions?

Answer 4:

So, FIXED and INCLUDE are both LOD expressions, and they're similar, but they're also different from one another.

FIXED expressions are like, "I don't care what I'm showing, I don't care what filters I'm applying, I want this calculated at this level."

So, if I say `{ FIXED [City] : AVG([Sales]) }` , I'm saying, "I want to find the average sales, and I want to find it at the city level, regardless of what other dimensions are in my view, regardless of what filters I'm applying."

INCLUDE expressions are like, "I want to find the level that I'm currently showing, and then also include this extra dimension in the calculation."

So, if I'm showing a view like this, and I want to find { INCLUDE [Product_Name] : AVG([Sales]) }, Tableau will take the average sales per product, and then find the average of those, and bring it up to the level I'm currently showing, which is Category.

The biggest difference between them is, FIXED expressions don't take into account dimension filters, while **INCLUDE** expressions do take into account what's happening in your view.

Practical Questions :

Task 1: Delivery Performance Classification

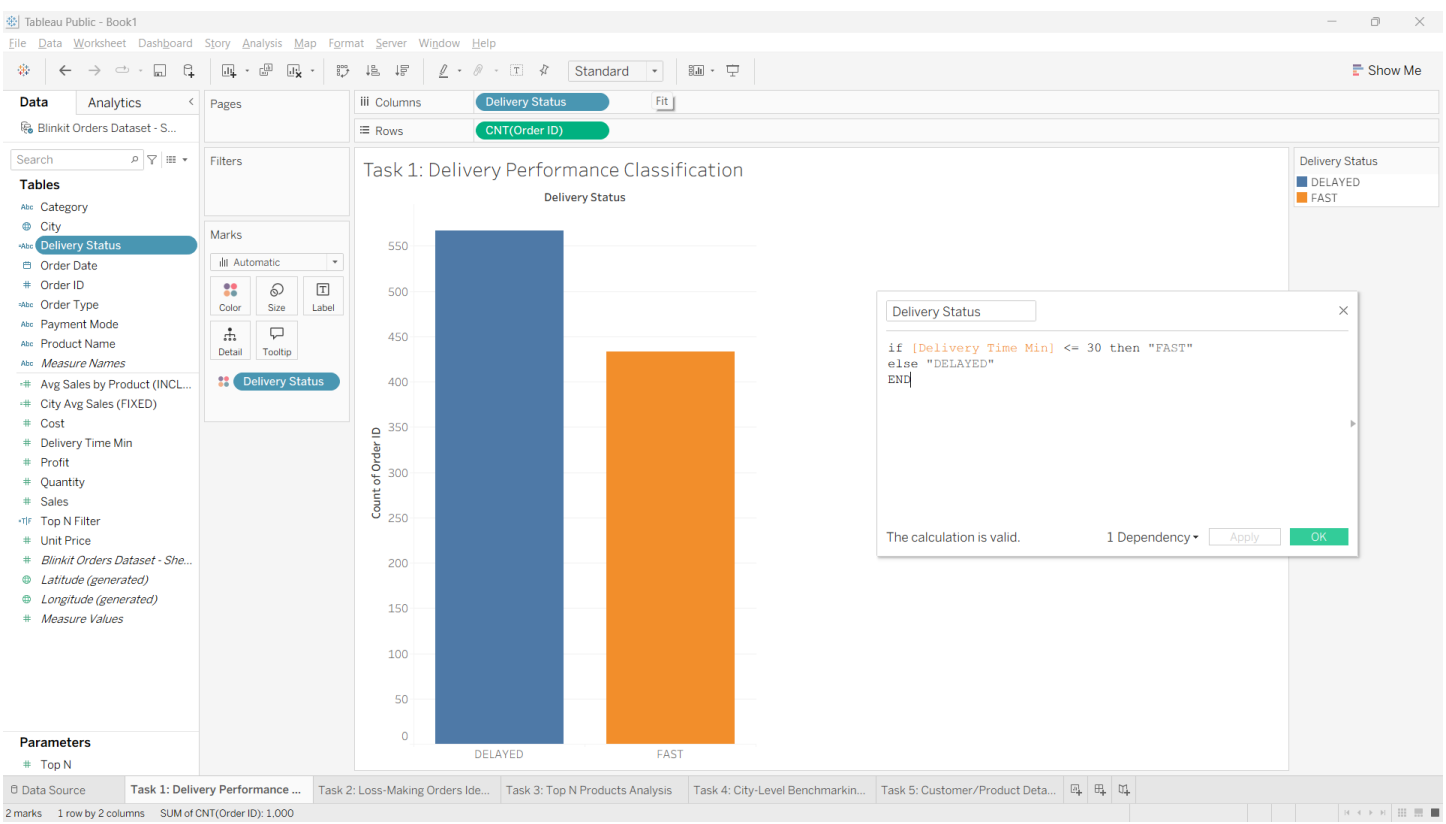
Goal: Label each order as **"Fast"** (≤30 min) or **"Delayed"** (>30 min)

Steps:

1. In Tableau, go to the **Data pane** (left side)
2. Click on **Analysis** menu (top) → **Create Calculated Field**
3. Name it: Delivery Status
4. Enter this formula:

```
IF [Delivery_Time_Min] <= 30 THEN "Fast"
ELSE "Delayed"
END
```

5. Click **OK**



Task 2: Loss-Making Orders Identification

Goal: Flag orders where **Profit < 0**

Steps:

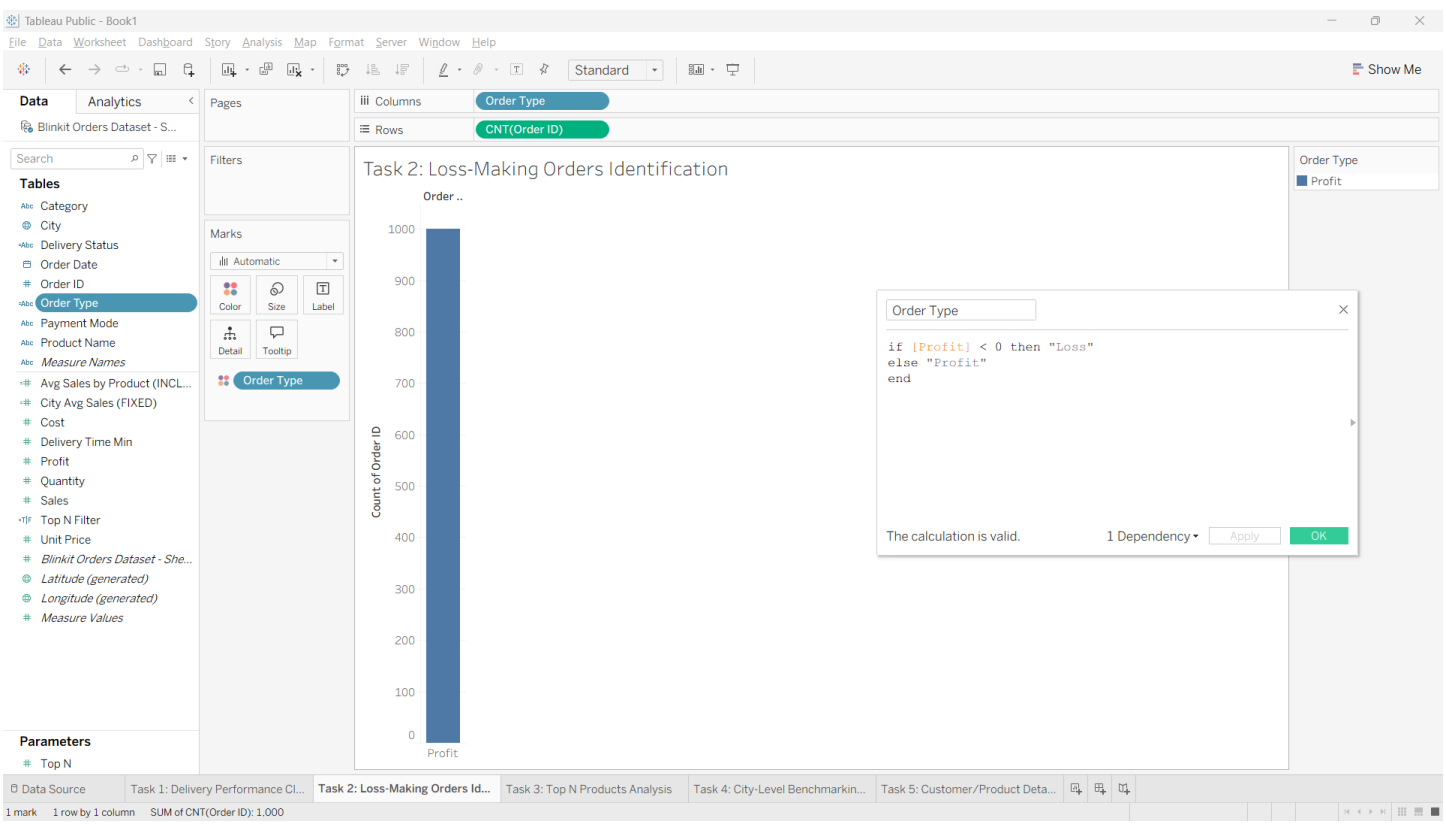
1. Create a new Calculated Field
2. Name it: Order Type
3. Formula:

IF [Profit] < 0 THEN "Loss"

ELSE "Profitable"

END

4. Click **OK**



Task 3: Top N Products Analysis (Parameter)

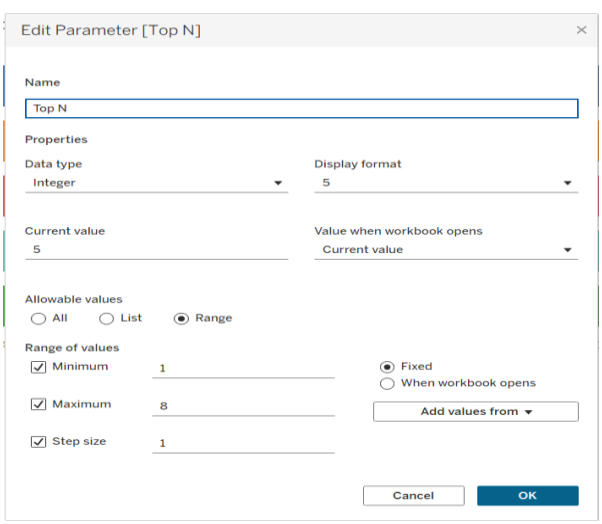
Goal: Let the user dynamically choose to see Top 5, Top 10, Top 15 products by Sales

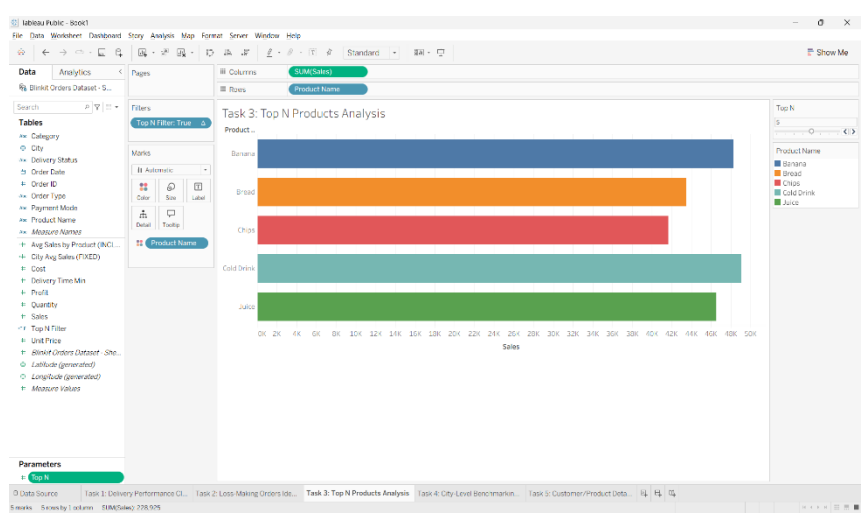
Steps — Part A: Create the Parameter

1. Right-click in the **Data pane** → **Create Parameter**
2. Settings:
    - Name: Top N
    - Data type: **Integer**
    - Current value: 5
    - Display format: Automatic
    - Allowable values: **Range** → Min: 1, Max: 20, Step: 1
3. Click **OK**

Steps — Part B: Create a Calculated Field using the Parameter

1. Create a new Calculated Field
2. Name it: Top N Filter
3. Formula:





RANK(SUM([Sales])) <= [Top N]

4. Click **OK**

Task 4: City-Level Benchmarking (FIXED LOD)

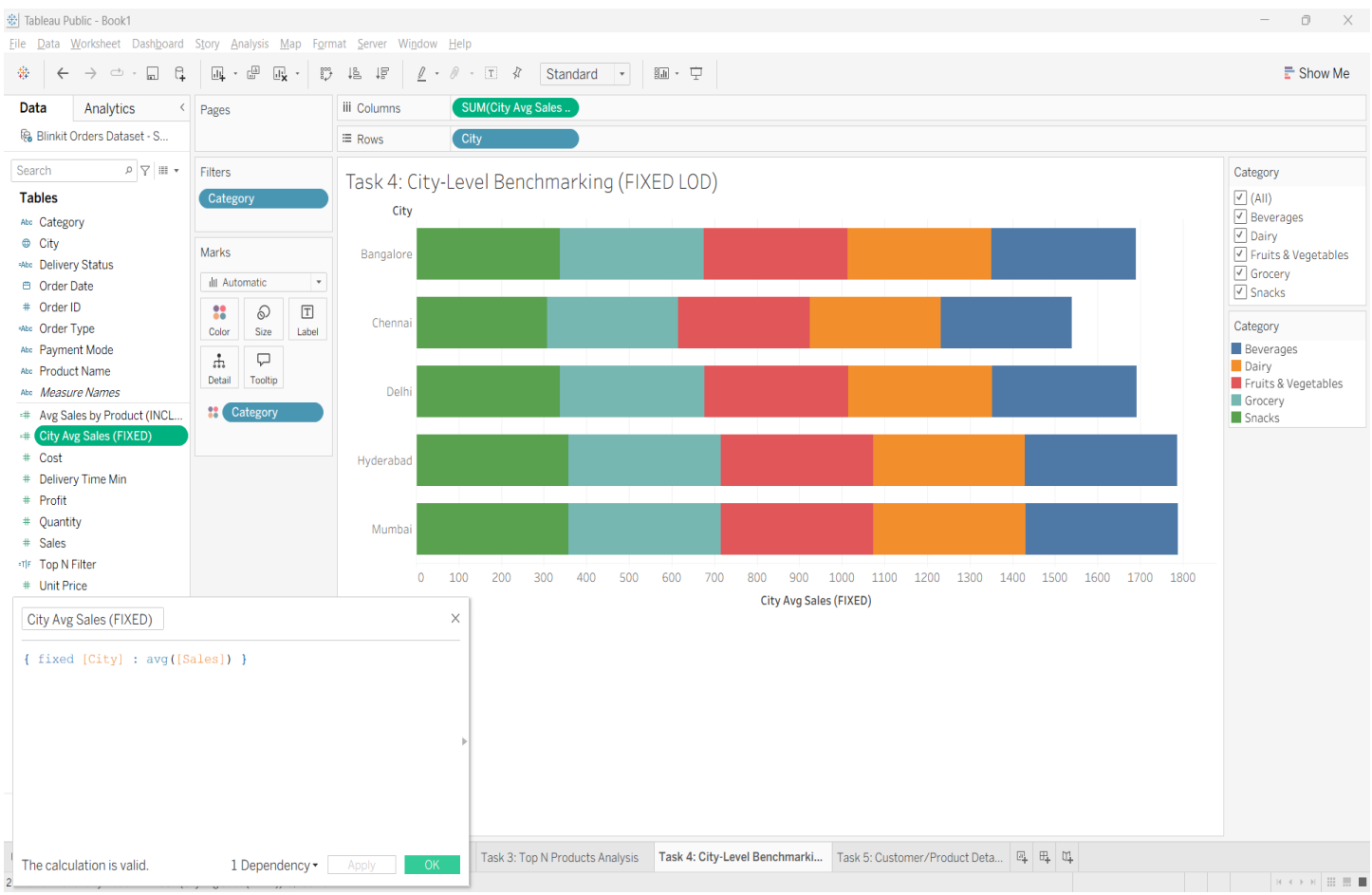
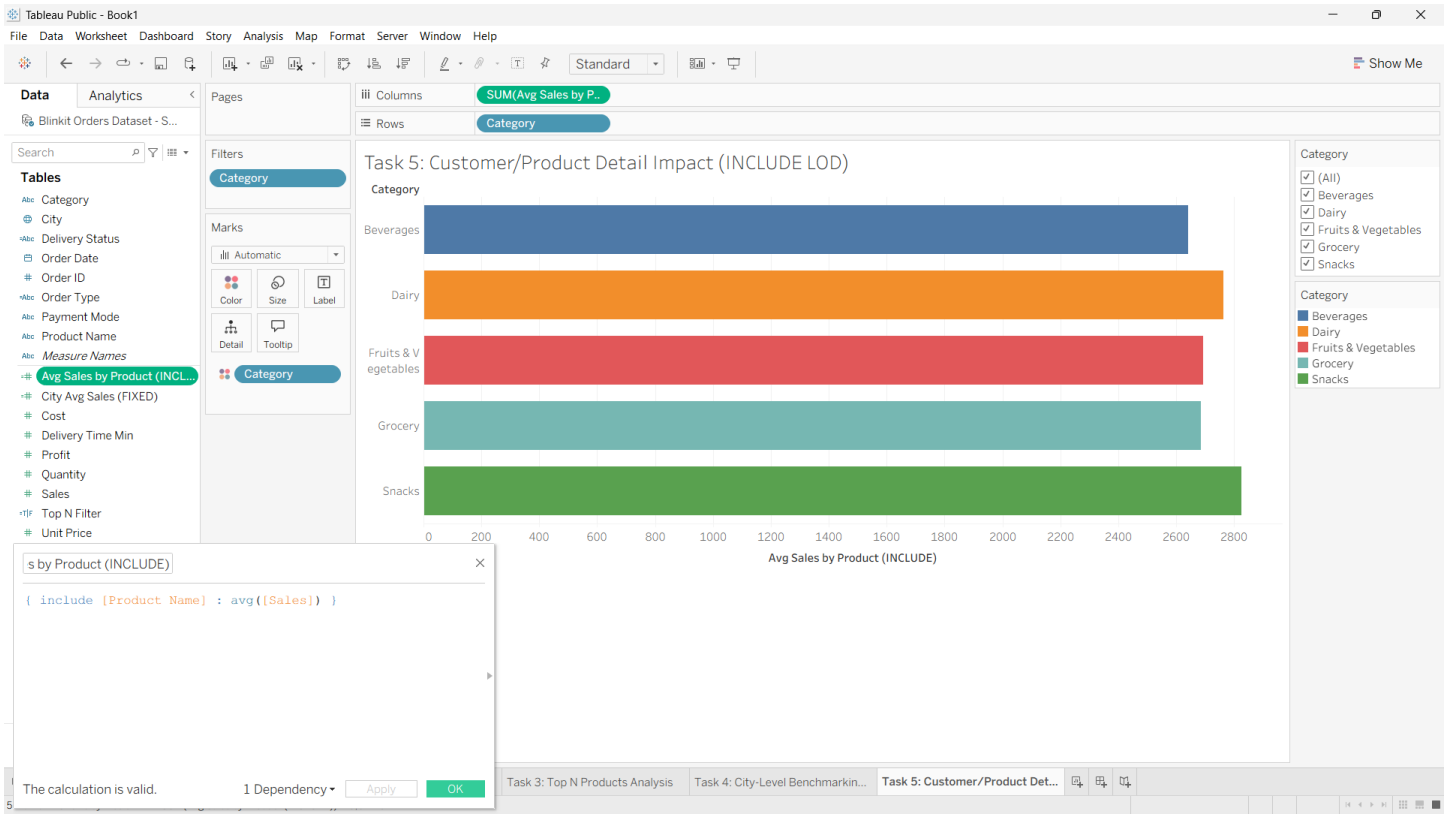
Goal: Calculate average sales per city — and this value should **NOT change** even if you apply a Category filter

Steps:

1. Create a new Calculated Field
2. Name it: City Avg Sales (FIXED)
3. Formula:

{ FIXED [City] : AVG([Sales]) }

4. Click **OK**



Task 5: Customer/Product Detail Impact (INCLUDE LOD)